

SkillChef — MVP Roadmap (28.05.2026 → 04.06.2026)

Stack: Next.js 14 (FE) · Java Spring Boot (BE) · PostgreSQL · Redis · S3/R2 · Flyway migrations
Team: 4 developers, 8 working days (weekends included) **Person → Dev mapping:** Person 1 = Марти (Auth/User) · Person 2 = Йосифов (Skill Tree/Learning) · Person 3 = Фичо (Challenges/AI) · Person 4 = Саши (Community/Infra)

Critical scope warning. The original task breakdown is sized for ~13 days. Fitting 04.06 requires cutting roughly 40% of scope. The plan below assumes the cuts listed in section 4 are accepted. If they aren't, 04.06 is not a deliverable date for this team — it's a slip risk.

1. Task schedule

Task ID	Task Name	Assignee	Depends On	Start	End	Duration (days)	Labels
T1	PostgreSQL + Flyway + USERS / PROFILE S7FOLLOWS schema	Марти	—	28.05	29.05	2	DB
T2	Next.js 14 skeleton + Tailwind + shared layout + minimum component library (Button, Card, Input, Avatar) + JWT auth context	Йосифов	—	28.05	29.05	2	FE
T3	DevOps base: Render web service config, env vars, Sentry. (PostHog, Logtail, Pingdom deferred.)	Саши	—	28.05	28.05	1	DevC
T4	Challenges/AI DB models: CHALLENGES, SUBMISSIONS, AI CONVERSATIONS	Фичо	T1 (in-flight)	28.05	29.05	2	DB

	AI_CONVERSATIONS, AI_MESSAGES						
T5	Storage layer: S3/R2 bucket, signed upload URLs, basic MIME/size validation, Java SDK. <i>(No image resize pipeline — defer.)</i>	Саши	T3	29.05	29.05	1	BE/In
T6	Skill Tree DB + seed: SKILL_DOMAINS, SKILL_NODES (tier, xp_reward), SKILL_PROGRESS, LESSONS, QUIZZES, XP_TRANSACTIONS. Seed only 2 domains × 4 nodes.	Йосифов	T1	30.05	30.05	1	DB
T7	Auth Service: signup, login, JWT issue/refresh, bcrypt hashing, Spring Security middleware. <i>(Email verification + password reset deferred.)</i>	Марти	T1	30.05	31.05	2	BE
T8	External integrations: OpenAI/Anthropic API client only. <i>(Mux, SendGrid deferred — videos hosted as raw URLs.)</i>	Фичо	T4	30.05	30.05	1	BE/In
T9	Recipe DB models: RECIPE_POSTS, COMMENTS, LIKES + indexes	Саши	T1, T5	30.05	30.05	1	DB
T10	User Service: profile CRUD, avatar upload (uses T5), public profile view, global_xp/level read.	Марти	T7, T5	01.06	01.06	1	BE

	<i>(Preferences deferred.)</i>						
T11	Skill Tree Service: fetch tree with progress overlay, unlock by tier, mark complete. <i>(Redis caching deferred.)</i>	Йосифов	T6, T7	31.05	01.06	2	BE
T12	Challenge Service + AI Service (sync only, no streaming, no prompt template system)	Фичо	T8, T7	31.05	01.06	2	BE
T13	Recipe Service + Engagement Service (likes, comments)	Саши	T9, T7	31.05	01.06	2	BE
T14	Lesson Service + Quiz Service + Progression Service (XP-to-level + level-up). <i>(Leaderboard deferred.)</i>	Йосифов	T11	02.06	02.06	1	BE
T15	Submission Service: media upload + sync AI scoring call + XP award. <i>(No background worker — synchronous request.)</i>	Фичо	T12, T5, T11	02.06	02.06	1	BE
T16	Social Service (follow/unfollow, lists, counts) + Feed Service (recency-only join on FOLLOWS)	Марти, Саши	T7, T13	02.06	02.06	1	BE
T17	FE: auth pages (signup/login) + profile page	Марти	T2, T10	03.06	03.06	1	FE
	FE: Skill Tree						

T18	visualization + lesson player + quiz UI + XP/level widget	Йосифов	T2, T14	03.06	03.06	1	FE
T19	FE: Challenges list/detail + submission upload UI + AI chat (sync)	Фичо	T2, T15	03.06	03.06	1	FE
T20	FE: Community feed + recipe detail + recipe creation page	Саши	T2, T13, T16	03.06	03.06	1	FE
T21	E2E integration + bug bash + production deploy on Render (<i>no custom domain — use *.onrender.com</i>)	All 4	T17, T18, T19, T20	04.06	04.06	1	QA/£

2. Dependency summary & critical path

Critical path (8 days, fills the window exactly):

T1 (2d) → T7 Auth (2d) → T10 User Svc (1d) → T16 Social+Feed (1d) → T20 Community FE (1d) → T21 Integration+Deploy (1d) = **8 days**.

Three parallel near-critical chains all converge into T21:

- **Learning:** T1 → T6 → T11 → T14 → T18 FE
- **Challenges/AI:** T1 → T4 → T8 → T12 → T15 → T19 FE
- **Community/Storage:** T3 → T5 → T9 → T13 → T20 FE

Hard handoffs (every one must hit its date or the deploy slips):

- T1 ends **29.05** → unblocks all backend services.
- T2 ends **29.05** → unblocks all four FE pages on Day 7.
- T5 ends **29.05** → unblocks avatar/recipe/submission media.
- T7 ends **31.05** → unblocks every protected endpoint.

There is **no buffer** anywhere. Every task is on the critical path or one day away from it.

3. Mermaid Gantt chart

Mermaid render error: Cannot read properties of undefined (reading 'type')

4. Required scope cuts (assumed in this plan)

These features from the original breakdown are **dropped or deferred** to make 04.06 fit. Treat them as post-launch backlog:

- **Email verification + password reset** (T7 simplified to login/signup only)
- **SendGrid integration entirely** (no transactional emails, no welcome email, no challenge reminders, no weekly digest)
- **Mux/Cloudflare Stream** (lesson videos stored as raw URLs; no transcoding, no signed playback)
- **Background worker / Spring Batch / Quartz** (submissions process synchronously — slow but works for low traffic)
- **AI streaming chat** (sync request/response only)
- **Prompt template system + token/cost tracking** (single hardcoded prompt per use case)
- **Redis caching layer** (direct DB reads — fine for MVP traffic)
- **Trending feed sort** (recency only)
- **Leaderboard query + XP transaction history endpoint**
- **Notifications system entirely** (in-app + email both dropped)
- **Image resize/optimization pipeline** (uploaded images served as-is)
- **Custom domain + Cloudflare DNS + SSL** (use `*.onrender.com` for MVP — buy domain in week 2)
- **User preferences** (no theme/notifications/language settings)
- **PostHog analytics, Logtail structured logging, Pingdom uptime** (Sentry only)
- **Skill tree seed reduced** to 2 domains × 4 nodes (vs. full 9-domain content)

5. Risk flags

Schedule risks

- **Zero buffer, zero margin.** Every task is on or adjacent to the critical path. Any 1-day slip on T1, T2, T5, T7, T13, or T20 pushes the deploy past 04.06.
- **One day for integration + deploy (T21).** A 4-service system with FE/BE/storage/AI touchpoints normally needs 2 days minimum. Expect to ship with known bugs.
- **Several 1-day tasks were 2 days in the original plan.** T10, T11, T13, T14, T15 are all compressed. If any one of them is actually 2 days, the chain collapses.

Architectural risks

- **Synchronous AI scoring (T15)** means slow submission UX — users wait 10–30s on the upload page while the AI call completes. Acceptable for demo, not for production traffic.
- **No background worker** means anything time-consuming (large image processing, AI calls, email sending) blocks request threads. Risk of timeouts under any concurrent load.
- **Марти schema gatekeeper bottleneck unchanged.** Day 1–2 PR reviews must be near-instant or DB tasks for Devs 2/3/4 stall.

Cross-team risks

- **Storage layer (T5) now has 0 buffer** — Саши has 1 day to ship signed URLs + Java SDK + validation. If it slips, T10, T13, T15 all slip and the whole plan collapses.
- **T16 is shared between Марти and Саши** on the same day. This requires tight coordination on the FOLLOWS join — recommend pair-programming on that 02.06.

Recommendations

- **Drop one of {Challenges, AI Chat, Community Feed}** and add the freed day as integration buffer. Each one is non-trivial and the demo doesn't need all three.
- **Demo prep should happen on 03.06 evening**, not 04.06 morning. If the deploy on Day 8 is also the demo day, plan a rehearsal the night before.
- **Push the 8-day deadline by 2 days (to 06.06)** if possible. The original 13-day plan is realistic; this 8-day plan is a gamble